Deva Sai Kumar Bheesetti

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Summary

Data Science and Data Engineering professional with 5 years of combined experience in software development, data automation, analytics, machine learning, and AI research. Skilled in building ETL pipelines, predictive models, dashboards, NLP workflows, vector search systems, and production-ready data applications using Python, SQL, PostgreSQL, MongoDB, PyTorch, scikit-learn, Tableau, Power BI, Docker, AWS, and GitHub Actions.

Education

University of Massachusetts Lowell — Lowell, MA
Master of Science in Computer Science, GPA: 3.94 — *Aug 2023 - Aug 2025*

- **Coursework**: Machine Learning, Natural Language Processing, Computation in Health and Medicine, Human-AI Interaction

K L University — India
Bachelor of Technology in Electronics and Communication Engineering, GPA: 3.52 — *2016 - 2020*

Work Experience

University of Massachusetts Lowell, Manning School of Business — Lowell, MA
Data Science Intern — *Oct 2025 - Present*

- Collected, cleaned, and analyzed LinkedIn profile, enrollment, and student engagement datasets to support academic research and institutional reporting.
- Performed exploratory data analysis, statistical analysis, and predictive modeling using Python, pandas, scikit-learn, Excel, and Jupyter Notebooks.
- Built dashboards and visual reports using Tableau, Power BI, Python, and Excel to communicate trends in student, alumni, and engagement outcomes.
- Created reproducible data workflows for cleaning, preprocessing, feature preparation, analysis, and stakeholder-ready reporting with light faculty mentorship.

University of Massachusetts Lowell, Department of Computer Science — Lowell, MA
Graduate Research Assistant, AI and Healthcare — *Jan 2024 - Aug 2025*

- Built an end-to-end NLP and machine learning pipeline over 2,636 de-identified opioid-related clinical notes to predict 15 Social Determinants of Health categories on an ordinal severity scale.
- Fine-tuned and evaluated Clinical Longformer, Longformer, DeepSeek-R1, and LLaMA 3.2 using PyTorch, Hugging Face, CUDA, and custom training workflows.
- Implemented multi-class classification, regression, and ordinal classification approaches with CORN heads and hybrid CEM-ORD plus CE loss functions for severity-aware prediction.
- Applied entropy-based active learning to select uncertain samples for physician review and improve rare-class calibration in imbalanced clinical data.
- Used Captum Integrated Gradients and LIME for token-level explainability, model interpretation, and clinician-facing review of informative clinical text features.
- Evaluated models against clinician annotations, achieving 82.4% exact match, 92.6% within-one ordinal accuracy, and a regression baseline MAE of 0.235.

University of Massachusetts Lowell, Manning School of Business — Lowell, MA
Graduate Assistant, Data Automation and Analytics — *May 2024 - Aug 2025*

- Automated reporting, grading, advising, web scraping, and event operations using Python, JavaScript, Google Apps Script, Docker, Selenium, Tableau, and Power BI, reducing manual workload by 70%.
- Built Tableau dashboards and Tableau Prep workflows for event attendance forecasting, data cleaning, business reporting, and operational planning for Manning Industry Days.
- Developed Python UnitTest scripts and deployed them on Linux servers with Docker to support automated grading for database-related coursework.
- Enhanced alumni data collection pipelines with Selenium, BeautifulSoup, and instruction fine-tuned LLM workflows for structured data extraction and cleanup.

OurFreedom.AI — New York, NY, Remote
Full-Stack Software Engineer, Data and Backend Systems — *Jan 2025 - Present*

- Built data-driven backend services using NestJS, TypeScript, MongoDB, and AWS, supporting social graph, user profile, moderation, reporting, and release workflows.
- Reduced profile and social graph load time by 60% by replacing repeated database calls with MongoDB aggregation pipelines, cross-collection joins, and targeted field projections.
- Designed reporting and moderation data flows for posts, profiles, and direct messages with validation, self-report prevention, admin logging, and audit-friendly review workflows.
- Standardized structured logging and environment-aware deployments across dev, UAT, and production using Docker, AWS ECS, GitHub Actions, and Keybase-managed environment files.

Infosys Ltd. Hyderabad, India
Senior Software Engineer, Module Lead, Client: Verizon *Aug 2022 - Aug 2023*
Software Engineer, Client: Verizon *Dec 2020 - Aug 2022*

- Built enterprise data synchronization and decisioning workflows for Verizon's offer management platform using REST APIs, Oracle SQL, rule-based processing, and scheduler-driven automation.
- Developed API logging utilities and Kibana reporting flows to improve visibility into custom and platform API traffic across production support workflows.
- Improved team throughput by 30% by automating offer creation, validation, migration, and data consistency checks across enterprise marketing systems.

Projects

- **Support RAG Evaluator**: Built a full-stack RAG evaluation platform for support-document retrieval, grounded answers, citations, refusal behavior, query logging, eval runs, analytics, streaming chat, OpenAPI docs, and pgvector search using NestJS, Next.js, PostgreSQL, Prisma, Docker, GitHub Actions, and Supabase. [Live] [GitHub]
- **VitaScan, AI Health Assistant**: Created a health data and AI platform with symptom intake, medical RAG, contextual chat, red-flag detection, health profiles, usage quotas, PostgreSQL, pgvector, Supabase RLS, OpenAI, Gemini, LangChain, NestJS, Next.js, and React Native. [Live] [GitHub]
- **Clinical Survival Analysis Dashboard**: Built a healthcare analytics dashboard using Python and Streamlit to analyze patient survival patterns with Kaplan-Meier curves, Cox regression, statistical testing, and clear visual outputs for clinical insight communication. [GitHub]
- **Cancer Subtype Classification**: Built a machine learning pipeline for breast cancer subtype classification using Python, scikit-learn, preprocessing, feature filtering, Logistic Regression, Random Forest, model evaluation, and Streamlit-based inference. [GitHub]

Skills

Programming and Data: Python, SQL, JavaScript, TypeScript, Java, Bash, pandas, NumPy, Excel, Jupyter Notebooks
Machine Learning and AI: scikit-learn, PyTorch, TensorFlow, Hugging Face, LangChain, XGBoost, LightGBM, CatBoost, RAG, NLP, LLMs, active learning
Data Engineering: ETL pipelines, data cleaning, preprocessing, web scraping, Selenium, BeautifulSoup, Google Apps Script, APIs, automation workflows
Databases and Analytics: PostgreSQL, pgvector, MongoDB, MySQL, Oracle SQL, Supabase, Prisma, Tableau, Power BI, Matplotlib, Seaborn, Plotly
Cloud and DevOps: AWS EC2, AWS ECS, S3, Docker, Kubernetes, GitHub Actions, Jenkins, Vercel, Render, Linux, Git